

Cloud Computing

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Defining a Cloud and Cloud Computing

Cloud computing is the the internet-centric way of doing computing.

Internet plays a fundamental role in cloud computing since it represents either the medium or the platform through which many cloud computing services are delivered and made accessible. on-demand delivery of computing services (like storage, servers, databases, networking, and software) over the internet, allowing users to access resources without owning or managing physical infrastructure. It provides scalability, flexibility, and cost efficiency by letting you pay only for what you use.

Three criteria to recognise whether a service is delivered in the cloud computing style:

1. The service is accessible over the internet
2. Zero capital expenditure to get started
3. Pay only for what you use as you use it

Cloud Computing

Cloud Computing refers to both the **computing environment** and **applications delivered as services** over the internet and **hardware and software** in the data centres that provide those services.

Cloud computing touches on the entire stack from the underlying hardware to the high-level software services and applications. It introduces the concept of everything as service mostly referred as XaaS, where the different components of a system can be delivered, measured and consequently priced as a service: Infrastructure, development platforms, databases, and so on.

Cloud computing is a technology in which we design computing systems, develop applications, leverage existing services for building software on the internet and also provide services using these resources.

Cloud Computing

Cloud Computing Characteristics, Features and Advantages:

- **On-demand self-service:** Users can provision resources (e.g., storage, compute power) instantly without human intervention.
- **Broad network access:** Services are available via the internet and accessible from multiple devices.
- **Resource pooling:** Providers serve multiple customers using shared infrastructure.
- **Rapid elasticity:** Resources can scale up or down quickly based on demand.
- **Use Anywhere:** Users access services based on a requirement without regard to where the services are hosted.
- **Pay-per-use:** Cloud computing allows renting infrastructure, runtime environment and services on pay-per-use basis. Economical.
- **Utility Computing:** This model has been referred to as utility computing due to the pay-per-use model
- **Dynamic Provisioning:** The use of dynamically provisioned IT resources constitutes a more attractive opportunity than buying additional infrastructure and software, whose sizing can be difficult to estimate and needs are limited in time.
- **Standardization:** With wide deployment of cloud computing systems, the foundation technologies and the systems enabling them are getting consolidated and standardized.
- **High level of Security:** Cloud Service Providers (CSPs) possess massive economies of scale, allowing them to invest vastly more resources into security infrastructure, expert personnel, and advanced threat intelligence than any typical single organization can afford on its own
- **Better CIA :** Confidentiality, Integrity and Availability.

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Cloud Computing

Cloud Computing Characteristics, Features and Advantages:

- **Benefits in performance** due to geographically distributed cloud environments and infrastructure (Edge Computing, Fog Computing, Regions, zones etc)
- **Simplified application acceleration** and Scalability, **Increased agility** in defining and structuring software systems
- **Energy Efficiency**
- Seamless creation and the use of **Third-party services** (AWS Marketplace)
- **Minimum maintenance**

History of Computing

- Mainframes
- PC
- Servers
- Clusters
- Grids
- **Data Centers:** Central location for computing resources—CPU, RAM, storage, and networking by connecting many computers together. Data centers provide resources for businesses' information technology needs: computing, storing, networking, and other services.
 - Lacks flexibility and agility and requires huge investment and maintenance costs. Often, building a new data center takes a long time and a big amount of money.

History of Computing

The Virtual Machine(VM): A VM is a **software-based computer**, composed of virtualized components of a shared physical computer—**CPU, RAM, HD and network**. It is a software-defined computer with a separate operating system and computing resources. The physical computer is called the **host machine** and virtual machines are **guest machines**. Virtual machines are abstracted from the computer hardware by a **hypervisor**.

With virtualization, a VM can be exported to a binary image file that can be deployed on any physical hardware at different locations. A running VM can be moved from one host to another.

The virtualization technologies virtualized physical hardware and caused a revolution in computer history, and also made cloud computing feasible.

History of Computing

- IBM was the first company to introduce the concept of virtual machines.
 - They developed the CP-40, an experimental time-sharing system, in 1964,
- While IBM pioneered the concept, VMware played a significant role in popularizing virtualization with the x86 architecture, enabling the creation of multiple virtual machines on a single PC
- The term **cloud** has historically been used in the telecommunication industry as an abstraction of the network in system diagrams. It then became a symbol of the most popular computer network: The internet.

Virtualization is a technology that uses software to create an abstracted, simulated layer over physical computer hardware, allowing a single physical machine to be divided into multiple virtual environments. This process enables components like processors, memory, storage, and networking to be split into multiple **Virtual Machines (VMs)**, each running its own independent operating system

Virtualization is a fundamental element of Cloud Computing. : The environments created by virtualization are called virtual because they simulate the interface that is expected by the user.

High-performance server can host several virtual machine instances, thus creating the opportunity of having customized software stack on demand. This is the base technology that enables Cloud Computing solutions, delivering virtual server on demands such as Amazon EC2, VMware vCloud and others. Together with hardware virtualization, *storage and network virtualization* complete the range of technologies for the emulation of IT structure.

Hypervisor

The *hypervisor* is a software component that creates and manages multiple virtual machines in a computer. It is a software layer that acts as an intermediary between the virtual machines and the underlying hardware or host operating system. The hypervisor coordinates access to the physical environment so that several virtual machines have access to their own share of physical resources.

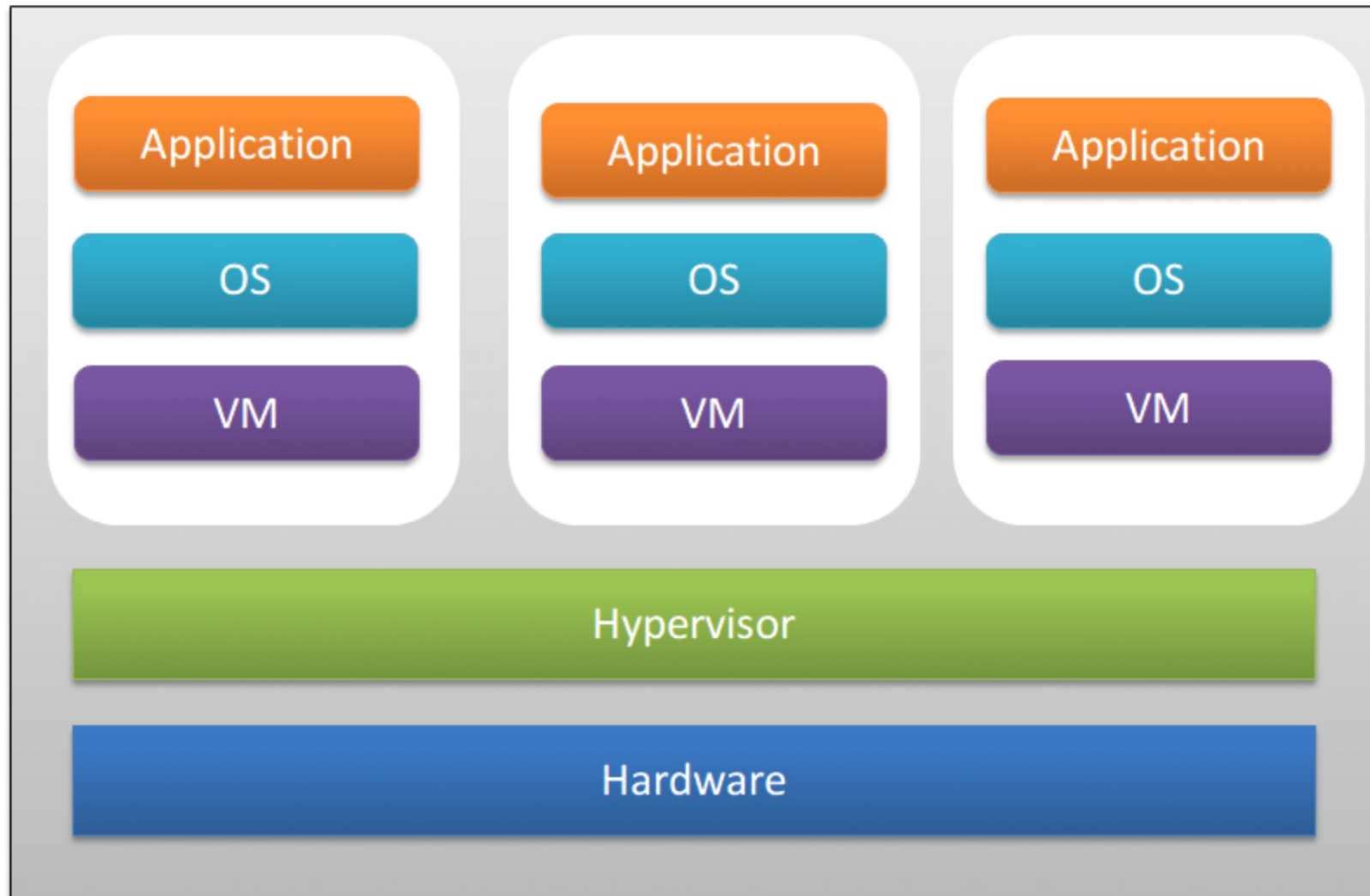
The following are the two main types of hypervisors.

Type 1 hypervisors

A type 1 hypervisor – also called a bare-metal hypervisor – runs directly on the computer hardware. It has some operating system capabilities and is highly efficient because it interacts directly with the physical resources.

Type 2 hypervisors

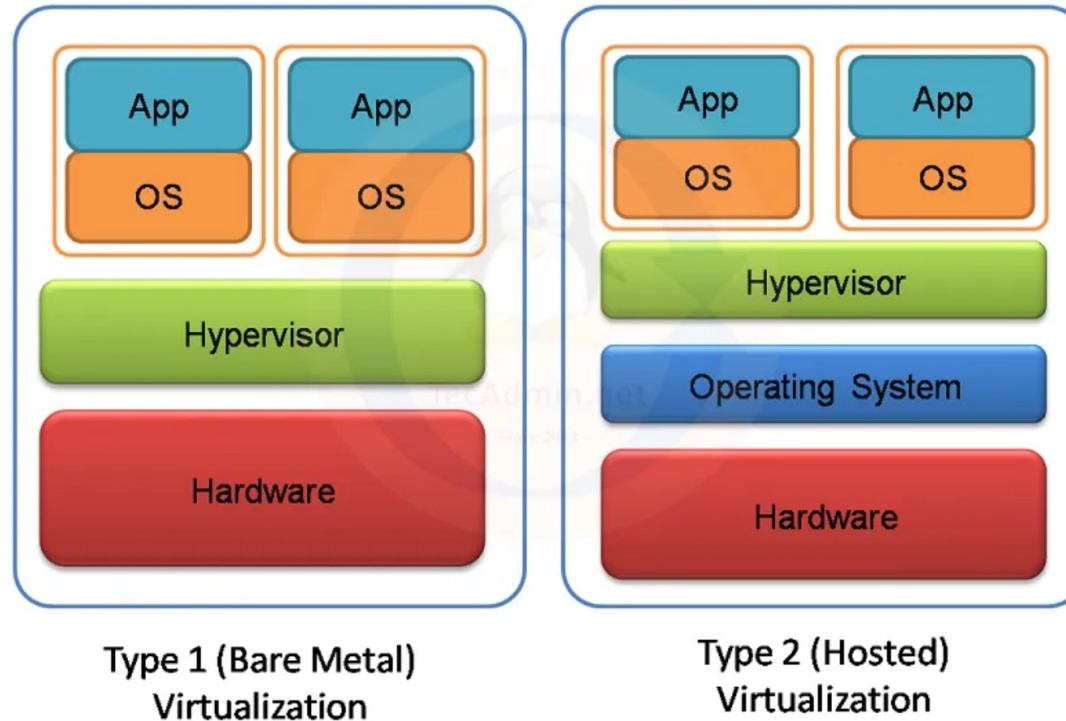
A type 2 hypervisor runs as an application on computer hardware with an existing operating system.



Understanding Hypervisors: Type-1 vs Type-2 and Full vs Para Virtualization

A **hypervisor**, also known as a **virtual machine monitor (VMM)**, is a software or firmware layer that enables multiple operating systems, known as guest operating systems, to run concurrently on a single physical host.

Type-1 Hypervisor vs Type-2 Hypervisor:



Type-1 Hypervisor (Bare Metal Hypervisor):

- Type-1 hypervisors run directly on the physical hardware without the need for a host operating system.
- They have direct access to the underlying hardware resources, offering better performance and efficiency compared to Type-2 hypervisors.
- Type-1 hypervisors are commonly used in enterprise data centers and cloud environments.
- **Examples include VMware vSphere/ESXi, Microsoft Hyper-V**

Type-2 Hypervisor (Hosted Hypervisor):

- Type-2 hypervisors run on top of a host operating system, which in turn runs on the physical hardware.
- They rely on the host operating system for managing hardware resources and providing device drivers.
- Type-2 hypervisors are typically used for desktop or development environments.
- Examples include Oracle VirtualBox, VMware Workstation, and Parallels Desktop.
- *The primary benefits of type-2 virtualization are its ease of use, cost-effectiveness, and compatibility*